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SYSTEM AND METHOD FOR A FLUID DISPERSAL CARTRIDGE

Abstract

A cartridge designed to dispense fluid via atomization is provided. The system generally comprises a cartridge, fluid, manifold, and air supply, wherein the air supply injects air through the manifold and into the cartridge through an air inlet of the cartridge. The air is manipulated by the cartridge in a way that creates a stream of fast-moving air above a fluid within the cartridge. This results in a zone of lower pressure above the fluid that results in the fluid to be suctioned into said stream of fast-moving air where it is atomized. The atomized fluid is then carried by the stream of fast-moving air out an atomization outlet where it is dispersed within the environment.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/311,530, filed on Feb. 18, 2022, which application is incorporated herein by reference.

FIELD OF DISCLOSURE

[0002] The subject matter of the present disclosure refers generally to a system and method for a cartridge designed to dispense fluid via atomization/diffusion.

BACKGROUND

[0003] Plant essences, botanical fluids, and other plant distillates, commonly called “essential oils,” are distilled fluids of plants, vegetables, nuts, seeds, roots, bark, flowers, etc. In some cases these distilled fluids can be made from non-organic substances as well but these will be included within the term “essence”, “botanical fluid”, “essential oil” or just “fluid” for sake of simplicity as well as “isolates” will also be included within these terms. These botanical fluids and essences typically have medicinal and/or therapeutic properties in addition to their valuable aromas that can be used in combination to create desirable fragrances. Unfortunately, the extraction of botanical fluids and essences from source plants is sometimes complicated, and comparatively expensive, based on the cost per unit volume of the botanical fluid and essence. As a result, colognes, perfumes, olfactory displays, and the like often use high rates of diluents with essences to reduce cost. They also may incorporate synthetic fluids, oils and artificial scents that may not replicate the comforting, familiar, and natural aroma of the natural essence. And most importantly, do not contain the beneficial medicinal and/or therapeutic properties the botanical fluids and essences possess due to their complex organic chemistry.

[0004] Therefore, there has been interest in ways to efficiently disperse essences in a way that best keep their beneficial bio-chemical properties.

[0005] Evaporation rates or atomization rates of essences are often inadequate though, meaning that in order to provide a controllable, sustainable, and sufficient amount of said botanical fluids in the surrounding environment to achieve a desired effect, one must provide some sort of mechanism to increase the amount in the environment. Therefore, one of the most common methods of atomization, wicking diffusers, often prove inadequate since they possess no air movement mechanism. As a result, other forms of diffusion have become increasingly popular for dispensing botanicals throughout an environment. These methods of diffusion include ultrasonic diffusers, oil lamps, candle diffusers, and aroma heaters, which all work by heating a fluid so that a botanical is dispersed into the surrounding environment. Though these are all effective methods of diffusion, heat often destroys or at least changes the constitution of essences. Thus, these methods of diffusion have limitations.

[0006] Further, hypersonic diffusers/atomizers work by diffusing water particles into the surrounding environment, wherein said water particles then carry the botanical fluids into the surrounding environment. This results in the dilution of the botanical fluids in addition to increasing the humidity of the environment, which may not be a desired feature. To make matters worse, water, botanical fluids, and/or essences diffused by hypersonic diffusers often damage surrounding equipment, furniture, and objects within the environment. Moreover, the “spitting” of botanical fluids and/or essences into the surrounding environment by hypersonic diffusers results in comparatively large droplets of said botanical fluids and/or essences, which can not only be particularly devastating to finishes of furniture but also wastes a substantial fraction of the botanical fluids and/or essences, driving up the cost to the user of said hypersonic diffuser.

[0007] Accordingly, there is a need in the art for a system that distributes botanical fluids and essences into a surrounding environment more efficiently than current methods can achieve.

SUMMARY

[0008] A system and method for a cartridge used to distribute a fluid via cold-air atomization is provided. In one aspect, the system distributes fluid into its surrounding environment via a jet of air. In another aspect, the system holds a fluid designed in a position such that it is pulled into the atmosphere by a jet of air. Generally, air from an air supply enters an air inlet of a cartridge, wherein it is manipulated by the cartridge in a way that creates a stream of fast-moving air above a fluid within the cartridge, which causes the fluid within the to be suctioned into said stream of fast-moving air where it is atomized. The atomized fluid is then carried by the stream of fast-moving air out the atomization outlet where it is dispersed within the environment. Therefore, the system of the present embodiment atomizes fluid into the surrounding environment without the need of heat or a carrier fluid such as water.

[0009] The system generally comprises a fluid, cartridge, manifold, and air supply, wherein the air supply injects air through the manifold and/or cartridge in a way that causes the fluid to atomize. The cartridge comprises a hard casing having an air inlet, atomization outlet, fluid reservoir, vacuum channel, and air duct. The manifold comprises at least one attachment point for removably securing the cartridge thereto. An air outlet of the manifold is configured to align with the air inlet of the cartridge. The air supply may secure to the manifold in a way such that it may provide air to the cartridge through the manifold. As air is pushed through the air duct of the cartridge by the air supply, a choke point within the air duct causes the air to become of a stream of fast-moving air. Because the choke point is located directly in front of the point in which the air duct meets the vacuum channel, a zone of reduced pressure is created above the vacuum channel, resulting in the atomized.

[0010] The foregoing summary has outlined some features of the system and method of the present disclosure so that those skilled in the pertinent art may better understand the detailed description that follows. Additional features that form the subject of the claims will be described hereinafter. Those skilled in the pertinent art should appreciate that they can readily utilize these features for designing or modifying other structures for carrying out the same purpose of the system and method disclosed herein. Those skilled in the pertinent art should also realize that such equivalent designs or modifications do not depart from the scope of the system and method of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] These and other features, aspects, and advantages of the present disclosure will become better understood with regard to the following description, appended claims, and accompanying drawings where:

[0012] FIG. 1 is a perspective view of a system embodying features consistent with the principles of the present disclosure.

[0013] FIG. 2 is a perspective view of a cartridge embodying features consistent with the principles of the present disclosure.

[0014] FIG. 3 is a front view of a cartridge embodying features consistent with the principles of the present disclosure.

[0015] FIG. 4 is a back view of a cartridge embodying features consistent with the principles of the present disclosure.

[0016] FIG. 5 is a left-side view of a cartridge embodying features consistent with the principles of the present disclosure.

[0017] FIG. **6** is a right-side view of a cartridge embodying features consistent with the principles of the present disclosure.

[0018] FIG. **7** is a top view of a cartridge embodying features consistent with the principles of the present disclosure.

[0019] FIG. **8** is a bottom view of a cartridge embodying features consistent with the principles of the present disclosure.

[0020] FIG. **9** illustrates how fluid and air flows through a cartridge embodying features consistent with the principles of the present disclosure.

[0021] FIG. **10** is a flow chart illustrating certain method steps of a method embodying features consistent with the principles of the present disclosure.

[0022] FIG. **11** is a flow chart illustrating certain method steps of a method embodying features consistent with the principles of the present disclosure.

DETAILED DESCRIPTION

[0023] In the Summary above and in this Detailed Description, and the claims below, and in the accompanying drawings, reference is made to particular features, including method steps, of the invention. It is to be understood that the disclosure of the invention in this specification includes all possible combinations of such particular features. For instance, where a particular feature is disclosed in the context of a particular aspect or embodiment of the invention, or a particular claim, that feature can also be used, to the extent possible, in combination with/or in the context of other particular aspects of the embodiments of the invention, and in the invention generally.

[0024] The term “comprises” and grammatical equivalents thereof are used herein to mean that other components, steps, etc. are optionally present. For instance, a system “comprising” components A, B, and C can contain only components A, B, and C, or can contain not only components A, B, and C, but also one or more other components. Where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where the context excludes that possibility), and the method can include one or more other steps which are carried out before any of the defined steps, between two of the defined steps, or after all the defined steps (except where the context excludes that possibility).

[0025] FIGS. **1-11** illustrate embodiments of a system **100** configured to atomize a fluid **106** into an environment using air **107**. FIG. **1** illustrates a perspective view of a system **100** designed to distribute a fluid **106** into a surrounding environment. FIG. **2** illustrates a perspective view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **3** illustrates a front view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **4** illustrates a back view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **5** illustrates a left-side view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **6** illustrates a right-side view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **7** illustrates a top view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **8** illustrates a bottom view of a cartridge **105** designed to distribute a fluid **106** into a surrounding environment. FIG. **9** illustrates the manner in which the cartridge **105** distributes fluid **106** into a surrounding environment. FIGS. **10** and **11** illustrate methods that may be carried out by a user using the cartridge **105**. It is understood that the various method steps associated with the methods of the present disclosure may be carried out as operations by the system **100** depicted in FIGS. **1-9**.

[0026] The system **100** generally comprises a fluid **106**, cartridge **105**, and air supply **115**, wherein the air supply **115** injects air **107** through the cartridge **105** in a way that causes the fluid **106** to atomize. In one preferred embodiment, as illustrated in FIG. **1**, the cartridge **105** is removably secured to a manifold **110** of the air supply **115** in a way that allows a user to remove a first cartridge **105** from the manifold **110** and secure a second cartridge **105** in its place, permitting a user to quickly replace cartridges **105** should the need arise. For instance, should a first cartridge **105** run out of fluid **106**, the user may obtain a second cartridge **105** containing a desired fluid **106**,

remove the first cartridge **105** from the manifold **110**, and secure the second cartridge **105** thereto. For instance, should the user desire a different fluid **106** to be atomized/nebulized/sprayed into the surrounding environment than that of the first cartridge **105**, the user may swap the first cartridge **105** having a first fluid **106** with a second cartridge **105** having a second fluid **106**.

[0027] As illustrated in FIGS. **1-9**, the cartridge **105** comprises a hard casing **105A** having an air inlet **105B** and an atomization outlet **105C**. In a preferred embodiment, the cartridge **105** is made of hard plastic, glass, enamel, or any combination thereof. However, other materials suitable for making a hard casing **105A** may be used without departing from the inventive subject matter described herein. As illustrated in FIG. **9**, a fluid reservoir **105D** within the hard casing **105A** is configured to hold the fluid **106** therein, and a vacuum channel **105E** connected to said fluid reservoir **105D** at a bottom connection point **105I** and an air duct **105F** at a top connection point **105J** allows a portion of the fluid **106** to encounter the stream of fast-moving air **107**. In a preferred embodiment, the air duct **105F** moves air **107** having an initial velocity from the base end of the cartridge **105** towards a choke point **105G** of the air duct **105F** located towards the top end of the cartridge **105**. The choke point **105G** of the air duct **105F** is a point in which the air duct **105F** narrows, resulting in a lower cross-sectional area of the air duct **105F**.

[0028] Additionally, in some preferred embodiments, as illustrated in FIG. **9**, the air duct **105F** is configured to expand past the choke point **105G** and top connection point **105J** until it reaches the atomization outlet **105C** and/or the outlet vent **105M**, which causes the stream of fast-moving air **107** to be directed towards the environment. However, in other preferred embodiments, the air duct **105F** may not extend past the choke point **105G** and top connection point **105J** but instead opens up into the fluid reservoir **105D**, allowing for atomized fluid to both collect within the fluid reservoir **105D** and move out of the atomization outlet **105C** and/or the outlet vent **105M**. Because the choke point **105G** is positioned directly upstream of the top connection point **105J**, as illustrated in FIG. **9**, the resulting Venturi effect causes the constricted air **107** to speed up as it reaches the top connection point **105J**, resulting in a drop in pressure over said vacuum channel **105E**. This drop in pressure causes the fluid **106** within the vacuum channel **105E** to be suctioned into the stream of fast-moving air **107** at the top connection point **105J**, where it is atomized and moved by the stream of fast-moving air **107** towards the atomization outlet **105C**.

[0029] In some preferred embodiments, the hard casing **105A** may further comprise an attachment element **105K**, knob **105L**, and outlet vent **105M**, wherein the attachment element **105K** and knob **105L** are configured to be used in conjunction with a manifold **110** while the outlet vent **105M** works in conjunction with the atomization outlet **105C**. The manifold **110** is configured in a way such that the air supply **115** attaches thereto so that the air supply **115** may supply air **107** to one or more cartridges **105** through said manifold **110**. As illustrated in FIG. **1**, a single manifold **110** may be used by an air supply **115** to provide air **107** to a single cartridge **105** or a plurality of cartridges **105** at once. A cartridge **105** may be secured to the manifold **110** using the attachment element **105K**, and the knob **105L** may be used to assist a user when removing a cartridge **105** from the manifold **110**. In a preferred embodiment, the attachment element **105K** is a locking tab and foot tab, as illustrated in FIGS. **1-9**, which secure the base end of the cartridge **105** to the manifold **110**. In a preferred embodiment, an air outlet of the manifold **110** is aligned with the air inlet **105B** of the cartridge **105** such that air **107** is supplied to the cartridge **105** from the air supply **115** and through said manifold **110** when the base end of the cartridge **105** is secured to said manifold **110**. The knob **105L** is preferably located on the top end of the hard casing **105A**, as illustrated in FIGS. **1-9**, in a way that assists a user to grip the cartridge **105** when removing it from the manifold **110**. This is necessary in instances when a cartridge **105** of a plurality of cartridges **105** secured to a manifold **110** is difficult to grip due to said cartridge's **105** location within said plurality of cartridges **105**, as illustrated in FIG. **1**.

[0030] As illustrated in FIGS. **2** and **3**, an outlet vent **105M** may be configured to secure within the atomization outlet **105C**, changing the height and/or angle at which a fluid is diffused into a

surrounding environment. The length and/or angle of the outlet vent **105M** may vary based on the fluid **106** being dispersed by the cartridge **105**. For instance, a fluid **106** that is more easily atomized and diffused into the surrounding environment may have an outlet vent **105M** having a shorter length, wherein the shorter length only slightly increases the height in which the atomized fluid **108** is distributed. For instance, a fluid **106** that is less easily atomized and diffused into the surrounding environment may have an outlet vent **105M** having a longer length, wherein the longer length greatly increases the height in which the atomized fluid **108** is distributed. Some cartridges **105** may not need an outlet vent **105M** if the fluid **106** is exceptionally easy to diffuse into a surrounding environment. In some preferred embodiments of the cartridge **105**, the outlet vent **105M** may be incorporated into the hard casing **105A** of the cartridge **105**, moving the location of the atomization outlet **105C** to the top of said outlet vent **105M**.

[0031] The air supply **115** is configured to provide air **107** to the cartridge **105** so that the fluid **106** therein may be atomized and dispersed into the environment. Types of air supplies **115** that may be used by the system **100** include, but are not limited to, an air pump, air compressor, compressed air canister, or any combination thereof. In a preferred embodiment, an air pump is used to push air **107** through a cartridge **105** and/or manifold **110**. In some preferred embodiments, the air pump may be secured to the manifold and/or cartridge via tubing. But in a preferred embodiment, the air pump may be incorporated into the manifold and/or cartridge that eliminates the need for tubing, creating a system with less parts that may be less prone to failure or necessitate cleaning less frequently. Additionally, filters of the air pump may be used to prevent the buildup of particulates in the manifold and/or cartridge, further eliminating the need for frequent cleaning. Types of air pumps that may be used by the system **100** include, but are not limited to, reciprocating pumps and rotary vane pumps. A switch of the pump may allow a user to activate the pump. In some preferred embodiments, one or more secondary switches may allow a user to control output of the pump, causing the pump to increase or decrease the flow of air **107** moving through the cartridge **105** and/or manifold **110**.

[0032] The system **100** may comprise a power supply. The power supply may be any source of power that provides the air supply **115** with electricity. In one preferred embodiment, the system **100** may comprise of multiple power supplies that may provide power to the system **100** in different circumstances. For instance, the system **100** may be directly plugged into a stationary power outlet, which may provide power to the system **100** so long as it remains within a certain distance of said stationary power supply. However, the system **100** may also be connected to a mobile power supply, such as a battery, so that the system **100** may receive power even when the system **100** is not connected to a stationary power outlet. In this way, the system **100** may always receive power so that a user may atomize a fluid **106** regardless of the location.

[0033] The fluid **106** contained within the cartridge **105** is preferably a natural, concentrated, liquid aromatic or medicinal fluid with or without aroma, such as cannabidiol, essences, botanical fluids, essential oils, and terpenes. The fluid **106** may be injected into the fluid reservoir **105D** via a reservoir hole **105H** that may be accessed via the atomization outlet **105C**, as illustrated in FIG. 9. Additionally, the reservoir hole **105H** allows atomized fluid **108** that collects within the air duct **105F** and around atomization outlet **105C** to flow back into reservoir in way such that it does not block the air duct **105F**. Therefore, in some preferred embodiments, the cartridge **105** may be refilled by the user once the fluid **106** within the reservoir is spent, and the cartridge **105** is self-cleaning in that air **107** moving through the cartridge **105** prevents debris from entering the atomization outlet **105C** while recondensed atomized fluid **108** has at least one path that allows it to flow back to the fluid reservoir **105D**. As such, a single cartridge **105** may be used to atomize a plurality of fluid **106s** and/or mixture of fluid **106s** without departing from the inventive subject matter herein.

[0034] FIG. 10 provides a flow chart illustrating certain, preferred method steps that may be used to carry out the method of swapping cartridges **105** to obtain the desired fluid **106** and then

operating the system **100** to atomize the desired fluid **106**. Step **1005** indicates the beginning of the method. During step **1010**, the user may acquire a system **100** comprising a first cartridge **105**, manifold **110**, and air supply **115**. During step **1015**, the user may perform a query to determine if the first fluid **106** within the first cartridge **105** is the desired fluid **106**. Based on the results of the query, the user may take an action during step **1020**. If the user determines that the first fluid **106** within the first cartridge **105** is the desired fluid **106**, the user may proceed to step **1040**. If the user determines that the first fluid **106** is not the desired fluid **106**, the user may obtain a second cartridge **105** having a second fluid **106** therein during step **1025**, wherein the second fluid **106** is the desired fluid **106**.

[0035] Once the user has acquired the second cartridge **105**, the user may remove the first cartridge **105** from the manifold **110** during step **1030**. In a preferred embodiment, the user must manipulate an attachment element **105K** and knob **105L** to remove the first cartridge **105** from the manifold **110**. The user may then attach the second cartridge **105** to the manifold **110** in place of the first cartridge **105** during step **1035**. The user may then perform a query to determine whether to turn on the air supply **115** during step **1040**, wherein turning on the air supply **115** will cause the system **100** to atomize the desired fluid **106**. Based on the results of the query, the user may perform an action during step **1045**. If the user determines they do not want to atomize the desired fluid **106**, the user may proceed to terminate method step **1055**. If the user determines that they would like to atomize the desired fluid **106**, the user may engage a switch of the air supply **115** that will cause said air supply **115** to draw power from the power supply and push air **107** through the cartridge **105** during step **1050**, resulting in the desired fluid **106** becoming an atomized fluid **108**. Once the air supply **115** has been turned on and the desired fluid **106** has become an atomized fluid **108**, the method may proceed to terminate step **1055**.

[0036] FIG. **11** provides a flow chart illustrating certain, preferred method steps that may be used to carry out the method of refilling spent cartridges **105** with a desired fluid **106**. Step **1105** indicates the beginning of the method. During step **1110**, the user may acquire a system **100** comprising a first cartridge **105**, manifold **110**, and air supply **115**. During step **1115**, the user may remove the cartridge **105** from the manifold **110** and may subsequently perform a query to determine if there is a workable amount of desired fluid **106** within said cartridge **105** during step **1120**. A workable amount of desired fluid **106** may be defined as the minimum amount of fluid **106** within a cartridge **105** that can be used by the system **100** to create an atomized fluid **108** at a desired rate. Based on the results of the query, the user may perform an action during step **1125**. If the user determines that a workable amount of desired fluid **106** is contained within the fluid reservoir **105D** of the cartridge **105**, the user may proceed to terminate method step **1140**. If the user determines that there is not a workable amount of desired fluid **106** within the fluid reservoir **105D**, the user may obtain a quantity of desired fluid **106** during step **1130**. The user may then add said quantity of desired fluid **106** to said cartridge **105** via the reservoir hole **105H** and the atomization outlet **105C** until a workable amount of desired fluid **106** is contained within the cartridge **105** during step **1135**. Once the user has filled the cartridge **105** with a workable amount of desired fluid **106**, the method may proceed to terminate method step **1140**.

[0037] The implementations set forth in the foregoing description do not represent all implementations consistent with the subject matter described herein. Instead, they are merely some examples consistent with aspects related to the described subject matter. Although a few variations have been described in detail above, other modifications or additions are possible. In particular, further features and/or variations can be provided in addition to those set forth herein. For instance, the implementations described above can be directed to various combinations and subcombinations of the disclosed features and/or combinations and subcombinations of several further features disclosed above. In addition, the logic flow depicted in the accompanying figures and/or described herein do not necessarily require the particular order shown, or sequential order, to achieve desirable results. It will be readily understood to those skilled in the art that various other changes

in the details, materials, and arrangements of the parts and method stages which have been described and illustrated in order to explain the nature of this inventive subject matter can be made without departing from the principles and scope of the inventive subject matter.

Claims

1-20. (canceled)

21. A system configured to atomize/nebulize/spray a fluid with a stream of air, comprising: a casing having a fluid reservoir, a vacuum channel, and an air duct therein, the air duct having an air inlet and an air outlet, wherein the fluid reservoir intersects the vacuum channel at a bottom connection point of the vacuum channel, the air duct intersects the vacuum channel at a top connection point of the vacuum channel, a narrowing of the air duct is located between the air inlet and the top connection point, and the air outlet is an atomization outlet.

22. The system of claim 21, further comprising a knob located on a top end of the casing.

23. The system of claim 21, further comprising: a manifold, having a manifold air outlet, an air supply, connected to the manifold, wherein the manifold is configured to receive the air from the air supply, and the manifold is configured to deliver the air to the air inlet from the manifold air outlet of the.

24. A system configured to atomize fluid into an environment, comprising: a manifold, a plurality of the systems of claim 21, with each system being a cartridge attached to the manifold, and an air supply, connected to the manifold, wherein the manifold is configured to receive the air from the air supply, and the manifold is configured to deliver the air to the air inlet of each of the cartridges.

25. The system of claim 23, wherein the air supply is at least one of an air pump and a compressed air canister.

26. The system of claim 24, wherein the air supply is at least one of an air pump and a compressed air canister.

27. The system of claim 21, further comprising the fluid in the fluid reservoir.

28. The system of claim 23, further comprising the fluid in the fluid reservoir.

29. A system configured to atomize fluid into an environment, comprising: a manifold, a plurality of the systems of claim 21, with each system being a cartridge attached to the manifold, and each cartridge containing a different fluid, and an air supply, connected to the manifold, wherein the manifold is configured to receive the air from the air supply, and the manifold is configured to deliver the air to the air inlet of each of the cartridges.

30. A method of atomize/nebulize/spray a fluid with a stream of air with the system of claim 27, wherein the air is caused to form a first air stream which flows through the air duct and is converted into a second air stream at the narrowing of the air duct, the second air stream has a speed greater than the first air stream, and the fluid is suctioned from the vacuum channel and into the second air stream, atomizing the fluid, and the atomized fluid exists out the atomization outlet.

31. A method of atomize/nebulize/spray a fluid with a stream of air with the system of claim 28, wherein the air supply causes the air to form a first air stream which flows through the air duct and is converted into a second air stream at the narrowing of the air duct, the second air stream has a speed greater than the first air stream, and the fluid is suctioned from the vacuum channel and into the second air stream, atomizing the fluid, and the atomized fluid exists out the atomization outlet.

32. A method of atomize/nebulize/spray a fluid with a stream of air with the system of claim 29, wherein within at least one of the cartridges, the air supply causes the air to form a first air stream which flows through the air duct and is converted into a second air stream at the narrowing of the air duct, the second air stream has a speed greater than the first air stream, and the fluid is suctioned from the vacuum channel and into the second air stream, atomizing the fluid, and the atomized fluid exists out the atomization outlet of the at least one of the cartridges.

33. A system configured to atomize/nebulize/spray a fluid with a stream of air under a Venturi

effect comprising: a casing having a fluid reservoir, a vacuum channel, and an air duct therein, the air duct having an air inlet and an air outlet, wherein the fluid reservoir intersects the vacuum channel at a bottom connection point of the vacuum channel, the air duct intersects the vacuum channel at a top connection point of the vacuum channel, a narrowing of the air duct is located between the air inlet and the top connection point, the system is configured such that when the fluid is present in the fluid reservoir and when the air forms a first air stream which flows through the air duct and is converted into a second air stream at the narrowing of the air duct, the second air stream has a speed greater than the first air stream, and the fluid is suctioned from the vacuum channel and into the second air stream under the Venturi effect, atomizing the fluid, and the air outlet is an atomization outlet.
